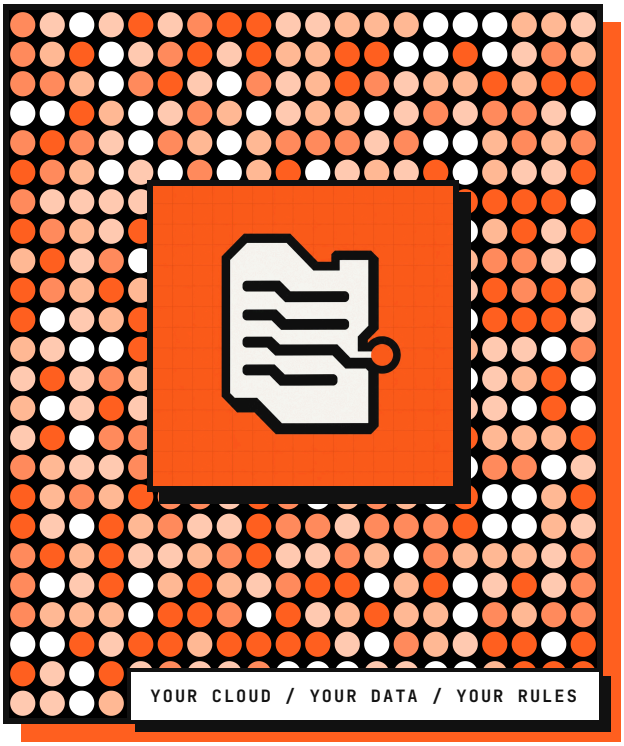
PRODUCT OVERVIEW

BACKOFFICE SUPER-INTELLIGENCE HARNESS.

PAPYRUS BY BEAG LABS · CUSTOMER-HOSTED SOFTWARE

A deployable AI execution platform that enables organizations to automate complex back-office operations across existing systems of record using configurable data models, durable agents, human approvals, and governed workflows.



THE PROBLEM

01
WORK LIVES BETWEEN SYSTEMS
Invoices, cases and approvals move across ERPs, inboxes, spreadsheets and terminals that were never designed to talk to each other.

02
AGENTS WITHOUT A BOUNDARY
Most AI tools act from a vendor cloud, outside your identity, data and approval controls, with no record of why they acted.

03
REPLACEMENT IS NOT AN OPTION
Systems of record carry years of configuration and policy. Rip-and-replace programs stall before they deliver.

WHAT PAPYRUS DOES

01
MODEL
Classes define your own record types: entities, vendors, cases, assets, mapped to the systems that hold them.

02
CONNECT
Dynamics 365, NetSuite, Odoo, SharePoint and mainframes stay authoritative. Papyrus reaches them through scoped connections.

03
EXECUTE
Durable agents and workflows run multi-step operations with checkpoints, retries, schedules and error handling.

04
GOVERN
Consequential actions pause for a human approval. Every action is recorded, so decisions can be reconstructed.

WHERE IT SITS

INPUTS
- ERP + finance systems
- Documents + email
- Forms + requests
- Legacy applications

→

PAPYRUS

DATA MODEL / APPROVALS / AUDIT

GOVERNED AGENT RUNTIME

Classes | Agents | Workflows | Apps + forms | Approvals

→

OUTPUTS
- Posted records
- Routed exceptions
- Internal apps
- Audit evidence

HOW IT WORKS

MODEL THE BUSINESS. RUN THE WORK.

Papyrus pairs a configurable data model with durable agents. Agents understand what a vendor, an entity or a case is in your organization, then carry the work across the systems that hold it, stopping for people where policy says they must.

THE PLATFORM

01 CLASSES

Your own record types with fields, relationships and source identities, created without code and kept in sync with connected systems.

02 RESOURCES

One catalog of every record, file and ID across connected systems, so people and agents find the same thing the same way.

03 FACTORY APPS

Describe a process and agents build the forms, dashboards and internal apps for it. Drafts publish only after review.

04 WORKFLOWS

A visual, top-to-bottom builder for triggers, branches, schedules and approvals. Agents can propose workflows for you to accept.

05 DURABLE AGENTS

Long-running sessions with memory, schedules and signals that survive restarts. Model-independent by design.

06 CONNECTORS

MCP-based, scoped tool access to external systems, with each connection tested and approved before it goes live.

GOVERNED EXECUTION LOOP



- **Agents propose. People approve.** External side effects cross an approval before they happen.
- **Your models, your gateways.** Point Papyrus at the model endpoints you have approved.
- **Durable by default.** Work checkpoints and resumes after failures or restarts.
- **Evidence built in.** Actions land in an append-only, hash-chained record.

BUILT FOR OPERATORS

01 ASK IN PLAIN WORDS

Staff start from Home and describe what they need. Papyrus finds the records, drafts the work and shows its sources.

02 EXCEPTIONS, NOT EVERYTHING

Routine steps run on their own. Only the cases that need judgment reach a person, with the evidence attached.

03 CHANGE WITHOUT A RELEASE

Add a field, a form or a workflow step in the product. No code deployment, no vendor ticket.

DEPLOYMENT + TRUST

RUNS INSIDE YOUR BOUNDARY.

Papyrus is customer-hosted software. It ships as a hardened container that runs in your own cloud account or data center, signs people in with your Microsoft Entra tenant, and keeps your records where they already are.

DEPLOYMENT OPTIONS

01
COMMERCIAL
Runs in your own cloud account.

- Deploys into your AWS or Azure account
- Sign-in through your own Microsoft Entra app
- Starts with a scoped pilot
- Updates delivered as signed images

02
GOVERNMENT CLOUD
Deployed with our team.

- GCC and GCC High Microsoft tenants
- AWS GovCloud or Azure Government
- Deployment profile set per environment
- Runbook and customer-controlled handoff

03
DOD + DISCONNECTED
A dedicated DoD container.

- DoD Microsoft cloud for sign-in and Graph
- Signed offline license, verified locally
- No callback to Beag Labs
- Same features as commercial

SECURITY CONTROLS

01
ENTRA-NATIVE IDENTITY
Microsoft Entra application roles are the only authority. No local accounts or separate role database.

02
SANDBOXED EXECUTION
Agent code runs in a network-denied sandbox with filesystem isolation. Secrets come only from vault or managed identity.

03
ACTION LEDGER + APPROVAL
Every external action moves from proposal to human approval to ledger to executor before it happens.

04
HASH-CHAINED AUDIT
Integration and action events form an append-only SHA-256 chain, ready for review and incident response.

05
SIGNED LICENSING
Licenses are signed with an HSM-held key and verified inside the deployment, online or fully offline.

06
HARDENED IMAGE
Built on Chainguard Wolfi, runs as non-root, ships with an SBOM, and is signed with Cosign on every release.

WORKS WITH WHAT YOU RUN

MICROSOFT ENTRA ID · MICROSOFT 365 · SHAREPOINT · TEAMS · EXCHANGE · DYNAMICS 365 · ORACLE NETSUITE · ODOO · MAINFRAMES + TERMINALS · SOAP / XML SERVICES · AWS + GOVCLOUD · AZURE + AZURE GOVERNMENT

ONE IMAGE

SAME PRODUCT, EVERY ENVIRONMENT

ONE CODEBASE. THREE WAYS TO RUN IT.

Commercial, government cloud and DoD deployments run the same Papyrus build. Only the identity cloud and the licensing change, so what you pilot is what you field.

Authorization note. Papyrus's identity, sandboxing, approval-ledger and audit design support a customer's own NIST CSF 2.0 and AI RMF programs. Papyrus is not, by itself, an authorization to operate or a claim of FedRAMP, DoD IL4/IL6 or other accreditation; risk assessment and authorization rest with the deploying organization.

IN PRACTICE

AUTOMATE WHAT HAPPENS BETWEEN SYSTEMS.

Start with one process that crosses systems and still depends on people moving data by hand. Papyrus takes the routine path end to end and hands people only the decisions.

ILLUSTRATIVE WORKFLOW: A MULTI-ENTITY INVOICE

01
INGEST
Receive the invoice and supporting evidence from email or upload.

02
RESOLVE
Link supplier and legal entities through Classes.

03
RECONCILE
Check allocations against purchase orders and source records.

04
APPROVE
Pause exceptions for a named approver, with evidence attached.

05
WRITE BACK
Post the authorized result into the original ERP.

WHERE TEAMS START

01
PAYABLES + RECEIVABLES
Invoice intake, matching, approvals and payment requests across entities.

02
DOCUMENTS + CASEWORK
Turn PDFs, forms and office files into structured, reviewable work.

03
LEGACY ACCESS
Give terminals and SOAP/XML services modern APIs and browser screens.

04
INTERNAL APPS
Forms, dashboards and trackers for the processes you already run.

THE FIRST 60–90 DAYS

FRAME
Confirm users, boundary, Entra tenant, deployment profile and acceptance criteria.

DELIVER
Model the Classes, connect the systems and build the workflow against real cases.

VALIDATE
Test identity, approvals, audit trail, failure handling and operator usability.

TRANSFER
Hand over the runbook, deployment assets and documentation to your team.

WHAT YOU GET

01
FEWER HANDOFFS
Routine steps run end to end without re-keying.

02
CLEAR TRACEABILITY
A record of every decision, change and approval.

03
YOUR STACK, KEPT
Systems of record stay in place and in charge.

04
ROOM TO GROW
Add Classes, apps and workflows without a rebuild.

NEXT STEP

PUT PAPYRUS ON A REAL WORKFLOW.

Talk to us about a pilot · james@beaglabs.com

BOOK A WALKTHROUGH →
beaglabs.com/contact